RATIONAL COHOMOLOGY ENDOMORPHISMS OF PRODUCT OF SPHERE WITH GRASSMANNIAN AND COINCIDENCE THEORY

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Abstract. We classified graded endomorphisms of the rational cohomology algebra of the product of a sphere and a complex Grassmannian, whose images are nonzero in the second cohomology of the Grassmannian.

We also derive necessary conditions for the generalized Dold spaces to satisfy the coincidence property, in particular the fixed-point property. As an application of our results, we obtain several sufficient conditions for the existence of a point of coincidence between a pair of continuous functions on certain generalized Dold spaces.

Proof. Let us consider the sum $\sum_{i=0}^{d} d_{2i}\lambda^i$, when λ is an integer. Clearly,

$$\sum_{i=0}^{d} d_{2i}\lambda^i \equiv 1 \pmod{\lambda}.$$

Hence, $\sum_{i=0}^{d} d_{2i}\lambda^i \neq 0$, if $\lambda \neq \pm 1$. When $\lambda = 1$, the sum is also positive and therefore nonzero. It remains to consider the case where $\lambda = -1$. Let $\chi(\mathbb{R}G_{n,k})$ denote the Euler-Poincaré characteristic of $\mathbb{R}G_{n,k}$ and be defined by

$$\chi(X) := \sum_{i \geq 0} \dim H^i(\mathbb{R}G_{n,k}; \mathbb{Z}_2)$$

where $\mathbb{R}G_{n,k}$ denotes the Grassmannian of real k-planes in $\mathbb{R}^n$. Now we observe that $\sum_{i=0}^{d} d_{2i}(-1)^i = \chi(\mathbb{R}G_{n,k})$ where $d_{2i} = \dim H^{2i}(\mathbb{C}G_{n,k}; \mathbb{Q}) = \dim H^i(\mathbb{R}G_{n,k}; \mathbb{Z}_2)$. It is a well known fact that $\chi(\mathbb{R}G_{n,k}) \neq 0$ if $k(n-k)$ is even.

Let us move to the other case where $\lambda \in \mathbb{Q}\setminus\mathbb{Z}$. Suppose $\sum_{i=0}^{d} d_{2i}\lambda^i = 0$ for some $\lambda = \frac{p}{q}$ where p and q are coprime integers. Since $d_0 = d_d = 1$, using the rational root theorem $p|1$ and $q|1$. Hence, $\lambda = \pm 1$, which is a contradiction. Therefore, we conclude that $\sum_{i=0}^{d} d_{2i}\lambda^i \neq 0$ for all $\lambda \in \mathbb{Q}$. □

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